

Md. Rifat Hossain

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EDUCATION

United International University <i>Bachelor of Science in Computer Science and Engineering</i> — CGPA: 3.87/4.00 Completed 125 out of 137 credits.	Dhaka, Bangladesh July 2022 – Present
Mirpur Cantonment School and College <i>Higher Secondary Certificate Examination</i> — GPA: 5.00/5.00	Dhaka, Bangladesh
Adamjee Cantonment Public School <i>Secondary School Certificate Examination</i> — GPA: 4.89/5.00	Dhaka, Bangladesh

EXPERIENCE

Undergraduate Teaching Assistant <i>United International University</i>	Mar. 2025 – Present Dhaka, Bangladesh
- Selected as an Undergraduate Teaching Assistant for three separate appointments: Mar. 2025–Jul. 2025, Aug. 2025–Oct. 2025, and Jul. 2026–Present. - Supported faculty members in conducting laboratory classes by assisting students with programming concepts, practical tasks, and course-related technical problems. - Provided individualized academic guidance and counseling, reviewed laboratory topics, and conducted supplementary tutorial sessions for students requiring additional support.	

PROJECT EXPERIENCE

LifeLens AI <i>React, FastAPI, MySQL, LangChain, AI/LLM</i> <i>Full-Stack & AI Developer</i> <i>Team Product</i> GitHub - Developed an AI-powered personal decision assistant integrating meal planning, productivity management, career and learning guidance, and financial planning into a unified platform. - Designed AI-assisted decision workflows that generate personalized recommendations while allowing users to modify suggested plans according to their preferences. - Implemented a FastAPI backend and React-based interface with MySQL data management and AI-driven conversational assistance.	
Sentinel Intelligence <i>YOLO, OpenCV, Python, Flask, ESP32, MySQL, IoT</i> <i>AI & IoT Developer</i> <i>Solo Product</i> GitHub - Built an AI-powered smart safety and emergency response system combining computer vision, IoT, embedded systems, and web technologies for automated hazard monitoring. - Implemented real-time YOLO-based fire detection and automated fire tracking with servo-controlled water nozzle positioning and dynamically controlled pump operation. - Integrated ESP32-based environmental monitoring using MQ-2, DHT11, and MPU6050 sensors for gas leakage, temperature, humidity, and seismic-event detection with automated ventilation and emergency responses. - Developed a Flask and MySQL-backed monitoring platform providing real-time safety status, remote device control, threshold configuration, historical data logging, event records, and visualization.	
People Flow Detection & Tracking <i>YOLOv8s, ByteTrack, OpenCV, Python, Supervision</i> <i>Computer Vision Developer</i> <i>Solo Product</i> GitHub - Developed a computer vision system for real-time people detection, persistent multi-object tracking, directional IN/OUT counting, and movement visualization. - Applied pretrained YOLOv8s for person detection and ByteTrack for maintaining persistent tracker identities across video frames. - Implemented line-crossing logic based on tracked center-point trajectories to count unique entry and exit events. - Generated movement-density heatmaps using OpenCV and produced annotated videos with bounding boxes, tracking IDs, and live IN/OUT counters.	
SyncUP <i>HTML/CSS, Node.js, Express.js, MySQL, Firebase</i>	

Full-Stack Developer | Team Product | [GitHub](#)

- Developed an integrated social media platform focused on user connectivity, interactive user-generated content, and personalized engagement.
- Implemented customized feeds, real-time location sharing, progress tracking, and personalized notifications for emergency and productivity-related alarms.
- Built the backend using Node.js and Express.js with MySQL and Firebase-based data and application services.

Football Game | Java, JavaFX, Object-Oriented Programming

Software Developer | Team Product | [GitHub](#)

- Developed a 2D multiplayer football game using JavaFX with competitive 1v1 matches, player management, ranking, and reward systems.
- Implemented interactive features including player trading and upgrades, in-game messaging, player marketplace functionality, and spinning rewards.
- Applied object-oriented programming principles to structure game entities, interactions, and gameplay functionality.

TECHNICAL SKILLS

Core Skills: Software Development, Full-Stack Development, REST API Development, Database Systems, Object-Oriented Programming, Data Structures & Algorithms, Artificial Intelligence, Machine Learning, Computer Vision, Data Analysis, Retrieval-Augmented Generation (RAG), Agentic AI, IoT, Embedded Systems, Web Scraping
Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL, HTML/CSS, Dart
Web Frameworks: React.js, Next.js, FastAPI, Node.js, Express.js, NestJS, Tailwind CSS
AI & ML: PyTorch, TensorFlow, Keras, Scikit-learn, YOLO, Ultralytics, OpenCV, LangChain, LangGraph, OpenAI, Hugging Face, spaCy, MLflow, NumPy, Pandas, Matplotlib, Seaborn
Database & ORM: MySQL, PostgreSQL, SQLAlchemy
Developer Tools: Git, GitHub, Docker, Jupyter Notebook, Google Colab, Google Cloud Platform, Postman, Swagger/OpenAPI
Others: Flutter, JavaFX, Firebase, Clerk, Better-Auth

HONOURS & AWARDS

UIU CSE Project Show	Jan. 2025
<i>1st Runner Up – Advanced Object-Oriented Programming</i>	<i>United International University</i>
UIU Merit-Based Scholarship	
<i>100% Merit Scholarship Holder – 5 times (4th, 5th, 6th, 10th & 12th Trimesters)</i>	<i>United International University</i>
<i>50% Merit Scholarship Holder – 2 times (7th & 9th Trimesters)</i>	
School Athletics	2018
<i>1st Prize – 100m & 200m Sprint</i>	<i>School Level</i>

VOLUNTEERING

Relief for Flood-Affected People	Aug. 2024
<i>United Nations Youth and Students Association of Bangladesh (UNYSAB)</i>	<i>Feni, Bangladesh</i>
• Participated in a UNYSAB relief mission supporting communities affected by the devastating floods in Feni.	
Eid for Street Children	Mar. 2024
<i>United Nations Youth and Students Association of Bangladesh (UNYSAB)</i>	<i>Dhaka, Bangladesh</i>
• Participated in a two-day community initiative providing iftar meals and support to underprivileged street children in Dhaka.	

LANGUAGES

Bangla: Mother Tongue
English: Fluent
Hindi: Speaking & Listening